

COMPUTING FUNDAMENTALS

USING COMPUTERS

Using a Workstation

- ✧ Follow documentation when installing computer equipment.
- ✧ Learn to navigate Windows interfaces

Using an OS

- ✧ Provides user interface, app platform, and system/file tools.
- ✧ Includes Windows, Apple Mac OS, Linux, Google Chrome, while *mobile devices* are supported by Windows Mobile, iOS, Android, Windows Mobile.
- ✧ **Virtualization** - allows running multiple OSs on one machine using a hypervisor.
- ✧ Learn to use File Explorer, Settings/Control Panel, & web browsers.

Managing an OS

- ✧ Use system tools for managing processes, memory, disks, and scheduling.
- ✧ A **Command Line Interface** allows control of the OS without a GUI. There may be more than one CLI included with an OS and the commands supported will vary.
- ✧ Installer usually becomes administrator; use standard accounts for daily tasks.

Troubleshooting and Support

- ✧ Learn the **CompTIA** troubleshooting steps.
- ✧ Identify external issues and understand risks like data loss or loss of service.
- ✧ Know how to find help.
- ✧ Use search engines effectively

UNDERSTANDING TECH BASICS

- **Technology Users** - Create content, Communicate, Manage and organize
- **IT Administrators** - Develop applications, Configure networks, Secure networks & Troubleshoot

Explore the Computing Cycle

- **CPU** - Executes instructions and processes data

- **RAM** - Temporarily stores information currently in use
- **Storage** - Stores programs and information long term

The Computing Cycle

- 1. Input** - This can be done manually or automatically using different devices.
- 2. Processing** - It is facilitated by the computer's CPU, which can process and manipulate data extremely quickly.
- 3. Storage** - When large amounts of data must be processed, the CPU relies on storage.
 - ★ **Short-term storage** - form of RAM
 - ★ **Long-term storage** - hard drive
- 4. Output** - Once the CPU manipulates the information and stores it, it then outputs the results.

Computing Devices

- ✿ **Computers** - was a term used to describe a person who was highly skilled at math. Now, a computer is a device that takes input and produces an output.
- ✿ **Desktop/PCs** - They tend to be bigger and easier to upgrade.
 - ★ **all-in-one computer** - computer hardware & video displays are integrated into one slim-line device. Can be difficult to upgrade and repair.
- ✿ **Notebook / Laptop** - Compact, portable computers with components in one case. Lightweight, energy-efficient, can run on AC power or DC batteries but is difficult to repair or upgrade.
- ✿ **Hybrid Laptop** - *2-in-1 laptops*, are designed to adapt to the user's needs. Some have removable keyboards & touchscreen displays.
- ✿ **Servers** - store information or provide network services to client computers. More advanced CPUs, RAM, & very large storage devices. The main difference between a regular computer & a server is their installed operating system.

✿ **Handhelds** - or tablets, designed to be powered by a DC battery and have OS specially designed for mobile applications.

- ★ **eReaders** - type of tablet with screens designed for reading and long battery life.

✿ **Gaming Device** - built for gaming, featuring advanced GPUs and very fast RAM. Optimized for high-performance graphics and have built-in networking capabilities.

✿ **Internet of Things (IoT)** - are physical devices embedded with sensors, software, and other technology that connect & exchange data with other devices and systems over the Internet or other networks. They can range from household devices to advanced industrial systems.

✿ **IoT Functions** - Collect information, Store information, Process information, Output information

DATA AND PRIVACY

☆ **Personally Identifiable Information (PII)** - Data is information being processed or saved in digital form on a computer.

☆ **Direct PII** - Information that can identify an individual

☆ **Indirect PII** - Info that is not unique to a single individual but is important to keep private.

☆ **Threat Actor** - Individuals or organizations who will use an individual's personal identifiable information with bad intent.

☆ **Identity Fraud** - When a threat actor uses someone's PII to steal their identity to acquire lines of credit, make purchases, or steal money from the victim's accounts.

How Companies Use Your Data

☆ **Targeted Marketing** - Presenting content for a specific user based on interests and demographics

☆ **Data Brokering** - Businesses that collect, sort, organize, and sell information.

☆ **Cookies** - Information that web servers send to a client's web browser and cookies are stored locally. Used to remember login info, preferences, viewed items, and user

interests. & used to enhance user experience on websites.

☆ **Bad Cookies** - Steal login credentials, Add malicious scripts to trusted sites, & Track users across multiple sites & build profiles.

Privacy in Communications

□ **Texting** - SMS Protocol. Sent through cell service in plaintext

□ **Instant Messaging (IM) Apps** - Sent through the internet, Some with end-to-end encryption

□ **Email** - Sends plaintext messages, Allows for encryption, one email at a time and FTP

□ **File Transfers** - File Transfer Protocol, use secure FTP (SFTP) for all file transfers.

□ **Privacy in Artificial Intelligence (AI)** - Internet as knowledge, AI and PII

Privacy Regulations

Federal

- ★ IPPA - Medical records
- ★ FCT Act - Unfair or deceptive marketing
- ★ FERPA - Student records privacy

State

- ★ CCA - California Consumer Privacy Act
- ★ New York SHIELD Act

European

- ★ GDPR - General Data Protection Regulation

Data Collection

Companies rely on digital data collection legally for marketing and service improvements.

□ **Analyzing data** - Using algorithms to extract patterns from data collected when you visit a site or fill out a form. Not performed by humans, algorithms are used for large scale analysis.

□ **Masking data** - The process of hiding sensitive information (PII) within a dataset to protect it from unauthorized access or exposure. This is typically done to ensure privacy and security.

□ **Monetizing data** - The process of turning data into a tangible asset that contributes to

the financial performance of a company. Ways to monetize data are by *selling data, targeted advertisements, licensing data to other companies, and data-driven decision making.*

Passwords and Encryption

- ❑ **Encryption** - the process of converting data into a code to prevent unauthorized access.
- ❑ **Algorithm** - A step-by-step procedure or formula for solving a problem. A set of well-defined instructions that take an input and produce an output after a finite number of steps.
- ❑ **Plaintext** - any text that has not been subjected to any form of encryption or encoding.

Encrypting Data

- ❑ **Encryption Methods**
 - ★ **Symmetric** - uses one key to encrypt and decrypt
 - ★ **Asymmetric** - uses different keys to encrypt and decrypt
- ❑ **Secure Protocols** - HTTPS, SSL/TLS, SSH
- ❑ **Virtual Private Network** - "tunneling"

Social Media Privacy

Leveraging social media platforms to share messages & images has become a popular method for staying connected with friends & family. Businesses often utilize the uploaded data to market their products. However, individuals with malicious intent may attempt to exploit site users.

Safety Tips

- ❖ Review your privacy settings
- ❖ Be mindful of what you share
- ❖ Clean up your friends list(s)
- ❖ Be cautious of quizzes & games
- ❖ Understand a platform's privacy policy
- ❖ Review your activity regularly

How Do You Create a Strong Password?

- ❖ **Authentication** - Factor(s) that prove you are who you claim to be.
- ❖ **Authentication Factors** - Something you are, have & know

- ❖ **Paraphrases** - Long, complex, memorable

Safety Tips

- ❑ Use a long password
- ❑ Include a mix of characters
- ❑ Avoid common words & phrases
- ❑ Do not use personal information
- ❑ unique password for each account
- ❑ Use passphrases
- ❑ Avoid predictable patterns
- ❑ Regularly update passwords

How Often Should You Change a Password?

- ❖ High-Security Accounts - **Every 60-90 days**
- ❖ Security Breach - **Immediately**
- ❖ General Accounts - **Every 6-12 months**
- ❖ Employer Accounts - **Company policy**

INTERNET TECHNOLOGIES

- ❖ **Internet** - A decentralized global computer network and with a common protocol suite
No single entity: Governs it, Manages it, Maintains it, Grows it

The **internet** started as an innovative project funded by the **U.S. Department of Defense** in the **1960s**. The goal was to create a strong and decentralized communication network that could still function even if parts of it were damaged—an important feature for military use. The **Defense Advanced Research Projects Agency** (DARPA) collaborated with **four universities** (UCLA, SRI, UCSB, & University of Utah) to build the network. This led to the creation of the **Advanced Research Projects Agency Network**

Web Pages and Websites

- ❖ **World Wide Web** - a collection of interconnected documents & resources accessed via web browsers. It uses the HTTP to facilitate communication between web servers & clients. Web resources identified by **URL**.

- ◇ **Hypertext Markup Language (HTML)** - Organizes everything you see on a web page
- ◇ **Hypertext Transfer Protocol (HTTP)** - The protocol that allows clients and servers to communicate
- ◇ **Web Server** - Software and hardware that use HTTP to respond to client requests made over the internet.

Basic Internet Terms

- ◇ **Protocols** - The internet relies on a set of rules and standards to ensure that data is transmitted accurately and efficiently.
- ◇ **Transmission control protocol (TCP)/ Internet protocol (IP)** - They allow different networks to interconnect and communicate, forming the basis of the modern internet.
- ◇ **Client** - device or software application that requests and uses resources or services
- ◇ **Server** - computer or software application that provides services or data to other computers
- ◇ **Client/Server Model** - a fundamental concept in networking, where clients and servers interact to perform various tasks.

Ways to stay safe while using the internet:

- ★ Use trusted engines
- ★ Verify URLs
- ★ Educate yourself
- ★ Avoid clicking on ads
- ★ Enable safe search features
- ★ Hover to preview
- ★ Use trusted sources
- ★ Use strong passwords
- ★ Keep browsers updated

Web Browsers - applications used to access web content

- ◇ Uses HTTP and HTTPS protocols
- ◇ Interprets HTML, CSS, and JavaScript to display: Text, Images, Multimedia & Interactive elements

Browser Extensions and Add-Ons

- ◇ **Add Functionality** - Extensions add specific capabilities or features.

- ◇ **Privacy Concerns** - Use caution with extensions that collect data.

- ◇ **Download from Trusted Sites Only** - Avoid downloading extensions from third-party sites.

Maintain Your Browser

- ★ Update your browser regularly
- ★ Clear the cache and cookies
- ★ Manage extensions and plugins
- ★ Use strong, unique passwords
- ★ Review privacy settings regularly
- ★ Backup bookmarks and settings
- ★ Be cautious about downloads

Internet Service Providers

a company or organization that provides internet access to users. ISPs own or lease infrastructure like cables, satellites, and servers.

ISPs offer subscription internet plans.

Plans differ based on bandwidth and speed.

ISPs provide: Internet access, IP addresses, Infrastructure, & Customer premises equipment (CPE)

Internet Connection Types

Wireless Internet

- ◇ **Cellular network** - mobile broadband

◇ **Fixed Wireless Internet** - using radio signals to provide internet access. Direct line of sight of the cell tower, Not affected by bad weather, Cost may be higher.

Cable Internet - Widely used in homes and small businesses. Most popular way to connect to the internet

Dial-Up - An older technology that uses telephone lines to connect to the internet. It is slow compared to modern options.

Digital Subscriber Line Internet -

Modem may need to be near a window
Uses existing telephone lines to provide internet access. It's faster than dial-up but slower than fiber-optic connections.

Fiber Optic - deliver extremely high-speed internet. It offers the fastest & most reliable connection. Uses light to carry information, EMI immunity, Longer lifespan than wire cabling

Satellite - Provides internet access via satellites. It is useful in remote or rural areas where other types of connections are unavailable. Bad weather can affect the signal

CYBERSECURITY

Practice of protecting systems, networks, and programs from digital attacks.

Cyberattacks usually aim to access, change, or destroy sensitive information or extort money from users

CIA Triad - Guide policies and practices to protect information and systems.

Confidentiality

- ◇ Ensuring that information is accessible only to those authorized to have access.
- ◇ Protect sensitive information from unauthorized access and disclosure.
- ◇ Methods: Encryption, access controls, authentication mechanisms, and data masking.

Integrity

- ◇ Ensuring the accuracy and completeness of data and that it has not been tampered with or altered.
- ◇ Maintain the trustworthiness and reliability of data.
- ◇ Methods: Hashing, digital signatures, checksums, and version control systems.

Availability

- ◇ Ensuring that information and resources are available to authorized users when needed.
- ◇ Ensure that systems and data are accessible and operational as required.
- ◇ Methods: Redundancy, failover mechanisms, regular maintenance, & robust disaster recovery plans.

Key components of Cybersecurity

- ◇ **Network security** - Protecting the integrity, confidentiality, and availability of data as it is transmitted across or within a network.
- ◇ **Information security** - Protecting the information stored in systems from unauthorized access, disclosure, alteration, and destruction.

◇ **Application security** - Ensuring that software and applications are secure from threats throughout their lifecycle.

◇ **Operational security** - Processes and decisions for handling & protecting data assets.

◇ **End-user education** - Training users to recognize and avoid potential threats, such as phishing emails and unsafe websites.

◇ **Disaster recovery and business continuity** - Strategies and plans for responding to and recovering from cybersecurity incidents.

Authentication

◇ the process of confirming the identity of a user, device, or entity in a computer system. It ensures that the entity requesting access is legitimate. One of the simplest forms of authentication is a username and password.

Multi-Factor Authentication (MFA)

◇ a security method that requires users to provide two or more types of verification to access a system, application, or data.

Most verification factors fall into these three categories:

- ◇ **Something You Know**: Information that the user knows, such as a password or PIN.
- ◇ **Something You Have**: A physical object that the user possesses.
- ◇ **Something You Are**: Biometric characteristics of the user.

Common Security Threat

- ◇ **Threat actor** - Individuals or entities that pose a potential risk to the security of an organization's information systems, networks, or data. They can vary widely in their motivations and attack methods. They are responsible for carrying out actions that can compromise the confidentiality, integrity, or availability of information.
- ◇ **Social engineering** - Manipulating individuals into divulging confidential information or performing actions through deception or psychological manipulation.
- ◇ **Lunchtime attack** - threat actor exploits the opportunity to gain access to applications or accounts a user is logged

into when an individual leaves their computer unlocked.

◇ **Shoulder surfing** - when a threat actor watches over the shoulder of an unsuspecting user to view their login credentials.

◇ **Tailgating** - tailgates an authorized employee, entering a secure building without the employee's knowledge.

◇ **Phishing** - Fraudulent attempts to steal sensitive information by pretending to be a trustworthy entity.

◇ **Smishing** - fraudsters send text messages to trick individuals into revealing sensitive information. SMS stands for Short Message Service.

Malicious Software (Malware)

any software intentionally designed to cause damage to a computer or computer network. Most rely on user interaction to become active. This activation can occur when a user *clicks on a link or opens a file*.

Common Malware Terms

◇ **Virus** - Programs that attach themselves to legitimate files & spread to other files.

◇ **Ransomware** - locks or encrypts data on a victim's system & demands payment to restore access.

◇ **Worm** - Standalone malware that replicates itself to spread to other computers. Do not need to attach themselves to an existing program.

◇ **Spyware** - secretly monitors user activity and collects personal information without the user's knowledge.

◇ **Adware** - automatically displays or downloads advertising material when a user is online.

◇ **Keyloggers** - Programs that record keystrokes made by a user, often to steal passwords and other sensitive information.

Security Tools

To safeguard against threats and ensure data integrity, confidentiality, and availability. These tools encompass sophisticated software applications and robust hardware devices.

◇ **Firewall** - A network security system that provides a barrier between trusted and untrusted networks, filtering incoming and outgoing traffic based on predefined security rules.

◇ **Proxy Server** - Intermediary servers that separate end users from the websites they browse. Depending on your needs, company policies, or privacy concerns. It provides varying levels of functionality, security, and privacy.

◇ **Antivirus software** - protect endpoints by preventing, detecting, quarantining, and removing malware.

◇ **Software patches** - Software and OS updates that fix performance bugs & address security vulnerabilities within a program to protect your system from cyber threats.

NETWORKING

Number Systems.

✧ **Decimal:** Used to describe data flow speeds or the amount of data storage.

✧ **Binary:** The fundamental language of computers, it is used to encode data & instructions.

✧ **Hexadecimal:** used in computing because it provides a format that can be more understood by people to represent binary values. *System: 1-9, A (10) - F*

Binary Conversion

✧ binary values - decimal values/decimal values - binary values is an important skill to master who plan to pursue a career in IT.

Decimal System (Base 10)

✧ the familiar numbering system used in our daily lives and also called "base 10"

✧ In base 10, each digit can be 0 to 9.

✧ The values of the places or positions of the numbers are based on powers of 10.

✧ Each number position is 10 times the value to the right of it

Binary System (Base 2)

✧ built around two unique digits: 0 and 1.

✧ The values of the places or positions of the numbers are based on powers of 2.

✧ Each number position is 2 times the value to the right of it

* **Conversion** (multi, add)

Binary to decimal
Hexa to decimal
Octal to decimal
(divide, multi, subtract)
Decimal to binary
Decimal to octal
Decimal to hexa

* **Grouping**

Binary to octal
Octal to binary
Binary to hexa
hexa to binary

Computer networks provide resource sharing and collaboration.

☆ **Resource sharing:** Networks allow us to share information and resources effectively. For example, printers and data storage can be accessed by multiple users across the network.

☆ **Collaboration:** Networks enable collaborative work. Whether working remotely or within the same building, networked computers provide ease of communication and cooperation.

Network

A computer network consists of interconnected devices known as **nodes**. These nodes can include computers, servers, printers, & other hardware. Networks can range from a simple connection between two computers to complex setups spanning vast geographical areas.

- **Network Media** - pathways that connect computers
- **Network Interface** - the point where two different systems interact
- **Network Port** - assign unique identifiers to different types of connection to keep interactions organized.
- **Networking Models:** Client -server & peer-to-peer

Common Networking Terminology

- **Node** - Devices like computers, servers, and printers connected to the network.
- **Host** - A client computer or a server.

○ **Transmission Media** - Physical pathways within a network that allow transmission of data from one device to another within a network

- **Protocols** - Rules and standards that govern data transmission, such as TCP/IP.
- **LAN** - covers a small geographic area.
- **WAN** - spans a large geographic area, often connecting multiple LANs.
- **PAN** - for personal devices, often connected using Bluetooth.
- **Port numbers** - uniquely identify connection endpoints and distinguish different services or processes running on a single device. For instance, **port 80** is commonly used for web servers.
- **Internet** - links million of smaller networks
- **VPNs** - secure connection between your device and the internet

Common Networking Devices

☆ **Switches** - devices that connect multiple devices within a LAN and use MAC addresses to forward data to the correct destination.

☆ **Routers** - devices that connect different networks together and route data packets between them. Essential for connecting a local network to the internet.

☆ **HUBs** - basic networking devices that connect multiple Ethernet devices, making them act as a single network segment. An older technology & have been replaced by switches.

☆ **Modems** - devices that modulate & demodulate signals for data transmission over telephone lines, cable systems, or satellite connections, enabling internet access.

☆ **Wireless Access Points (WAPs)** - devices that allow wireless devices to connect to a wired network using Wi-Fi.

☆ **Wireless Local Area Point (WLAN)** - uses wireless to connect devices within a limited area.

Network Addresses

Most modern networks rely on two widely used systems to connect devices:

- I. **Ethernet:** serves as the foundational connection system, including

physical components such as cables, NICs, & unique identifiers known as MAC addresses.

- II. **TCP/IP:** A suite of software protocols that enable devices to communicate seamlessly by speaking the same "language." IP addresses identify & route data between devices.

Media Access Control address

- * physical /burnt-in address
- * unique identifier assigned to a network IP address interface card (NIC) or network adapter. It helps identify devices on a local network. MAC addresses are globally unique, meaning that no two hardware components will have the same MAC address.
- * MAC addresses are 48 bits in length and are expressed using 12 hexadecimal numbers. Example: 0C-4F-97-1A-48-BD

Internet Protocol Address

- * an IP address is a unique numerical identifier assigned to each device connected to a network. Essential for identifying devices and facilitating the routing of data across networks, including the internet.
- * considered a logical address because it is assigned by the network administrator. Unlike the MAC address, which is permanent and unchangeable, an IP address can be part of a custom network design.
- * There are two IP address standards: IPv4 and IPv6.
 - o An IPv4 address is 32 bits long, expressed in decimal
 - o IPv6 address is 128 bits long, expressed in hexadecimal.

* **Dynamic Host Configuration Protocol (DHCP)** - Auto and static configuration

Why do we need two addresses?

- MAC addresses ensure accurate data delivery within a local network, while IP addresses enable data to be routed across different networks. Together, they provide a

comprehensive addressing system that supports both local and global communication.

Wired Network Connectors

- **Twisted pair** - pairs of wires twisted together to minimize crosstalk
- **Standards** - 568A, 568B
- **JR45** - Standardized connector

Wireless Network Standards

- Modulation - data riding a carrier
- 802.11 standards - compatibility
- Wireless security - encryption for data in transit

Wireless Network Speed

- Latency - lag time or delay
- Interference - other signals, physical interference
- Distance from WAP - Attenuation

Wireless Band Options

- 2.4 GHz - Longer range, Better through walls
- 5 GHz - Higher data speeds, Less crowded
- vHertz (Hz) - Measurement of cycles per second (e.g., wave or signal)

Virtualization on the Network

Virtualization - Computers in computers
Virtualization Requirements

- o Computer
- o Hypervisor or Virtual Machine (VM): Type I/II hypervisor
- o Guest Operating Systems

Hypervisors and Guest Operating Systems

□ **Hypervisor** - Software that manages hardware

- * **Type 1 Hypervisor** - Runs on Computer's hardware; bare metal installation
- * **Type 2 Hypervisor** - Runs on top of host operating system; opposite of bare metal

Hardware and Virtual Machines (VMS)

Container Virtualization

- * Portable software packages
- * Allows several apps to run simultaneously

Virtual Servers

- ✳ Software based server
- ✳ Allows multiple servers to run on a single physical machine

APPLICATIONS AND SOFTWARE

Software - set of instructions or programs that tell a computer how to perform specific tasks. Software provides step-by-step instructions for a computer to perform specific tasks.

Common Software Types

- ★ **Web Browser** - It enables the information you retrieve from the Internet to be displayed as text, images, links, and other elements.
- ★ **Email Software** - method of sending and receiving digital messages over the Internet. Allowing you to communicate with others using an email address.
- ★ **Messaging Software** - Text messaging is a way to send short written messages between mobile phones. Often called SMS and instant messaging is a way to send and receive real-time text messages over the Internet.
- ★ **Productivity Software** - facilitate the creation, editing, management, and sharing of information and tasks, thereby enhancing productivity and efficiency in various professional and personal activities.
- ★ **Collaboration Software** - Video conferencing allows people in different locations to see and talk to each other in real-time over the Internet. Online storage & file sharing is a service that saves files & shares them with others easily on the internet.
- ★ **Work Management Software** - typically includes features for task assignment, scheduling, progress tracking, collaboration, and reporting. Its primary goal is to improve productivity, streamline workflows, and ensure that projects are completed on time and within budget.
- ★ **Remote Assistance Software** - allows a user to receive support from a remote location. This software enables one computer to be accessed and controlled by

another computer over a network or the Internet.

★ **Word Processing Software** - application software designed for the creation, editing, formatting, and storing of data in tabular form.

★ **Spreadsheet Software** - designed for organizing, analyzing, and storing data in tabular form.

★ **Presentation Software** - used to create, edit, and display visual presentations.

★ **Diagramming & Graphics Software** - software tools designed to create, edit, and manage diagrams, and image and video editing applications are software tools designed to manipulate, enhance, and transform visual and audiovisual content.

Safety Managing Software

- **Trusted Site** - these sites are known for their reliability, adherence to legal standards, and protection against malware and other malicious threats.
- **App Store** - Platforms like the Apple App Store, Google Play Store, and Microsoft Store have stringent security measures and vetting processes for the software they offer.

Software Licensing

- **End User License Agreement (EULA)** - A legal contract between the software developer or vendor and the end user who purchases, installs, or uses the software. It outlines the terms and conditions under which the software can be used, including any usage, distribution, and modification restrictions.
- **Subscription Model** - A licensing and delivery method in which users pay a recurring fee to access and use the software.
- **Open-Source** - offers freedom, transparency, and community-driven development, allowing users to modify and share the software.

CompTIA Troubleshooting Method

1. Identify the problem
2. Establish a theory of probable cause
3. Test the theory

4. Establish a plan of action to resolve the issue
5. Implement the plan or escalate the issue
6. Confirm the solution works
7. Document findings

HARDWARE

Computer hardware refers to the physical parts of a computer, like the CPU, monitor, keyboard, and mouse.

Common Metric Prefixes

Base 10

□ based on numbers 0-9 (10 numbers total) in the number system. Storage drive manufacturers often use Base 10 when listing storage capacity.

Base 2

□ includes metric prefixes in Base 2, based on the binary numbering system of 1's and 0's. Base 2 uses powers of two

□ Operating Systems use Base 2 to calculate the size of data when it is stored on a disk or transmitted on a network.

Term	Represents (units) Base 2	Represents (units) Base 10
Kilo (K)	2^{10} or 1,024	10^3 or 1,000
Mega (M)	2^{20} or 1,048,576	10^6 or 1,000,000
Giga (G)	2^{30} or 1,073,741,824	10^9 or 1,000,000,000
Tera (T)	2^{40} or 1,099,511,627,776	10^{12} or 1,000,000,000,000
Peta (P)	2^{50} or 1,125,899,906,842,624	10^{15} or 1,000,000,000,000,000

Throughput and Speed

□ **Bandwidth** - the maximum amount of data that can be transmitted over a network

connection in a given amount of time. It is typically measured in bps, Kbps, Mbps, or Gbps.

□ **Throughput** - actual rate at which data is successfully transferred over the network. It is also measured in bps, Kbps, Mbps, or Gbps

□ **Hertz** - a unit of measurement that indicates how many times something happens per second.

Computer Hardware

□ **Form Factor** - shape and size of a device.

□ **Motherboard** - crucial component of a computer, also known as the mainboard or system board. It acts as the central hub that connects all the other components of the computer,

□ **Read Only Memory** - type of non-volatile memory used in computers and other electronic devices to store data that does not change, such as firmware or system software.

□ **BIOS/UEFI** - Firmware stored on a computer's motherboard that initializes and tests hardware during the startup process and provides runtime services for operating systems and programs.

- UEFI is a modern version that is replacing BIOS, providing more features.

□ **Central processing unit** - main part of a computer that performs most of the processing inside the computer.

□ **Memory (RAM)** - computer's short-term memory. Where it temporarily stores data that the CPU needs to access quickly while running programs.

□ **Graphics processing unit** - specialized computer chip designed to quickly render images and videos, making graphics look smooth and detailed.

□ **Network interface card** - allows a computer to connect to a network, enabling it to communicate with other devices.

□ **Volatile** - loses all stored data when the computer is turned off.

□ **Non-Volatile** - does not lose stored data when the computer is turned off.

Computer Storage

□ **Internal Storage (Hard Drive)** - used in computers to store and retrieve digital information, including the operating system, software applications, and user files.

- *Examples:* magnetic storage devices and solid-state storage devices.

□ **External Storage** - any data storage device that is not housed within the computer's main chassis and is connected externally.

- *Examples:* external hard drives (i.e., USB, eSATA, or Thunderbolt), USB flash drives, or memory cards.

□ **Optical Storage** - reads and writes data from optical discs, such as CDs, DVDs, and Blu-ray discs, using laser technology.

□ **Network Attached Storage (NAS)** - connects to a network and provides shared storage that multiple users and devices can access.

□ **Cloud Storage** - a service that allows you to save and access your data over the Internet instead of on a local drive. (Google Drive, Dropbox, and Onedrive).

□ **Storage Area Network (SAN)** - specialized network that provides fast, reliable access to large amounts of data storage.

Peripherals

□ A "*peripheral*" in computer hardware refers to any external device that connect to and works with the computer to add functionality.

□ *common peripherals:* keyboard, mouse, monitor, speakers, printers, scanners

Uninterruptible Power Supply (UPS)

device that provides backup power to a computer or other electronic equipment in case of a power outage, ensuring that the device continues to operate without interruption.

Installing Connected Devices

□ **Plug-on-Play:** a term in computing and electronics to describe a feature that allows a device to be connected to a computer or other system and be recognized and

configured automatically without requiring manual installation or configuration by the user.

□ **Driver:** specialized software program that allows the operating system and other software applications to communicate with hardware devices.

OPERATING SYSTEMS

A computer operating system (OS) is software that manages hardware and other software resources on a computer. It acts as an intermediary between users and the computer hardware.

Five Features an Operating System Provides.

1. Communicate with Hardware - An OS must effectively communicate with system hardware, and it accomplishes this through driver software.

2. User Interface - An OS must also offer a means for users to interact with the computer to issue commands or requests.

3. Provide a File System - The OS organizes files and directories, providing a structured way to store and retrieve data.

4. Application Platform - An OS provides a platform for programs by offering a stable and consistent environment in which applications can run.

5. Security - The OS enforces policies and access controls, ensuring that applications run in a secure environment. It manages user authentication and permissions, preventing unauthorized access to a system's resources.

Types of Operating Systems

✦ Embedded Operating System -

Software designed to manage hardware and software in a specialized device, like a smartwatch, cash registers, or car control system, often with limited resources and specific functions.

✦ **Open-Source** - Software whose source code is freely available for anyone to view, use, modify, and distribute.

♦ **Commercial Operating System** -

Software that is sold and licensed for use, typically developed by a company for profit.

♦ **Windows (Microsoft)** - offers *two primary types of operating systems*: Windows for personal computers (PCs) and Windows Server for servers.

♦ **MacOS (Apple)** - The Mac operating system, known as *macOS*, is a series of graphical operating systems developed and marketed by Apple Inc. for their line of Macintosh computers.

♦ **Linux** - Linux distributions, often referred to as *Linux distros*, are variations of the Linux operating system that bundle together the Linux kernel with other software, tools, and applications to create a complete operating system tailored for specific needs and preference.

User Interface

♦ **Text-Based or Command Line Interface (CLI)** - allows for precise control over the system. Users can execute complex commands and scripts to automate tasks, making it highly efficient for repetitive or batch operations. The commands are issued in text format.

♦ **Graphical User Interface** - They are generally more intuitive and easier to use, especially for beginners. They provide visual elements like icons, buttons, and menus that make navigation straightforward. The commands are issued by selecting icons.

Software Installation

♦ **Installing Software** - The goal of installation is to ensure that the software is properly integrated with the operating system and can function as intended.

♦ **Removing (Uninstalling) Software** - This process involves deleting the application's files, configurations, and any associated data from the system to free up resources and ensure that the software no longer affects the system's performance or functionality.

♦ **Mobile Applications** - Installing a mobile application is a straightforward process, but the steps can vary slightly depending on

whether you use an Android or an iOS device.

Computer Management

□ **Device** - A "*hardware device*" refers to any physical component that is a part of a computer system or connected to it.

□ **Driver** - A "*hardware driver*," is a specialized software program that allows an operating system to communicate with a specific hardware device. It acts as translators, converting the operating system's generic commands into device-specific instructions that the hardware can understand and execute.

□ **Startup Applications** - Software programs that are configured to automatically launch when a computer system boots up.

□ **Task Management (Manager)** - A system utility in Windows operating systems that provides detailed information about the computer's performance and any open applications, processes, and services.

□ **System Services** - Specialized programs or processes that run in the background of an operating system to perform essential functions and support the overall operation of a computer or network.

File Management

□ **Window File Explorer** - A file management application included with the Microsoft Windows operating system, and it is also known as *File Explorer*. It provides a GUI for accessing, managing, and organizing files and folders on a computer.

□ **Apple Finder** - The default file management application used in macOS, the operating system for Apple computers. It provides a GUI for accessing, managing, and organizing files, folders, and applications on a Mac.

□ **File Fragmentation** - It is when a file is broken into pieces and stored in different locations on a disk, slowing down access and performance.

□ **File Defragmentation** - A process of recognizing fragmented files on a disk to store them in contiguous sections, improving access speed and performance.

□ **File System Root** - The starting point of the file system on a given drive, from which all other directories and files branch out.

□ **Files and Folders** - The way data is stored and organized on a computer.

Data Backup

- copy of important files & information stored in a separate location to protect against data loss.

□ **Computer Personal Files** - The data and documents created, stored, and managed by individual users on their computing devices.

□ **System Files** - Essential files that are part of the operating system. These files are critical for the proper functioning and stability of the system.

□ **Local Backup** - It is the process of copying and storing data on a physical storage device that is located on-site or in close proximity to the source of the data. This can include external hard drives, USB flash drives, NAS or other local storage media.

□ **Cloud Backup** - The process of copying and storing data on remote servers hosted on the internet. This type of backup is managed by third-party service providers who offer storage space and management services.

DATABASES

A database is an organized storage system based on the idea of tables. Tables in a database can store any type of information.

Examples include: Names, Dates, Files, Pictures, Comments

Database Characteristics

□ **Centralized Access** - Multiple users can get into or access the database at the same time.

□ **Security** - Databases include security measures to ensure that only authorized users can access the data.

□ **Recovery** - Databases are easy to back up, so database administrators (DBAs) can restore the data if something goes wrong.

□ **Scalability** - Can easily handle a few records or millions of them.

Database

- ★ Quick access to billions of records
- ★ Data validation ensures accurate records
- ★ Thousands or millions of concurrent users

Spreadsheet

- ★ Simple structure
- ★ Great for small to medium datasets
- ★ Perfect for a single user or small group

Working with a Database

□ **Import Records** - Bringing data from an external file, such as a spreadsheet or form, into a database. It can bring in large amounts of information very quickly and accurately.

□ **Queries** - A search of the database. They help you find the data you need.

□ **Reports** - it is the formatted output of data, usually for analysis or presentation. They give you easy-to-read summaries of the data, which could be shown in graphs or charts.

□ **Offline Database** - can be used without an Internet connection. It reduces the risk of cyber-attacks.

□ **Online Database** - users can access over the Internet. This allows employees or customers of an organization to reach it from anywhere.

□ **Local Database** - loaded on a local server, and the server is placed on the company-owned network so employees at the office and those outside the local network can use it.

□ **Cloud Database** - putting the database & data online, hosted & managed by the cloud provider.

Database Structure

□ **Structured Database** - The data is organized in a fixed format, often in tables with rows and columns. This type of data is easy to search and analyze.

□ **Non-Structured Database** - The data is stored in a format that doesn't follow a strict

schema or structure, such as text, images, or videos. You can still find what you need, but there is no order.

▫ **Semi-Structured Database** - a mix of structured & non-structured data. It doesn't fit neatly into tables but has some organizational properties, like tags or markers, to separate elements.

Types of Databases

○ **Relational Database** - Stores and organizes data in tables. It uses relationships between tables to link data, making it easy to retrieve and manage related information.

○ **Non-Relational Databases** - You can organize and access the items based on their type or label, but there's no fixed structure like shelves or drawers.

○ **Key-Value Database** - type of non-relational database that stores data as pairs of keys & values.

○ **Document Database** - Stores data in flexible documents, like company inventory or blog posts. It can store multiple pieces of information in one field.

Relational Database

○ **Database Schema** - The architecture of the databases' structure and organization.

○ **Concept Data Model** - A simple way to organize and define the data needed for a system with no real technical details.

○ **Logical Data Models** - It is more detailed and explains how to structure the data. It includes technical details like primary keys, foreign keys, entity names, entity relationships, and attributes. *However, it doesn't show how to implement the database with a specific database management system.*

○ **Physical Data Models** - The most detailed and shows exactly how to set up a database using a specific database management system. It includes technical details and performance needs for the hardware that will run the database.

○ **Tables in a Relational Database** - Tables form the structure of a database. Tables consist of records and fields.

○ **Primary Keys** - A special identifier that makes each record unique. A social security number (SSN) can be a primary key, as no two SSNs are alike.

○ **Foreign Keys** - Helps link data from one table to another. A Foreign key in one table references the primary key from a different table.

Database Concepts

○ **CRUD**: Basic operations on data – Create, Read, Update, Delete.

○ **Data Definition Language (DDL)**: Commands for defining database structure – create, alter, drop

○ **Data Manipulation Language (DML)**: Commands to manipulate data. Insert, select update, delete.

○ **Relational Database Management System (RDBMS)**: Software managing relational databases; uses tables, keys, and SQL queries. (MySQL, Oracle, PostgreSQL)

CODING

Coding is writing instructions for a computer to follow, like a recipe to bake a cake.

What is a programming language?

☆ A programming language is a way to write code that computers can understand and run.

☆ Programming languages have a strict syntax, or a set of rules for the language.

Programming Language Terms

☆ **Machine Language** - The most basic programming language, consisting of binary code that a computer's CPU can directly execute.

☆ **Syntax** - Sets the rules for how to write the language. These rules include statements, symbols, capitalization, punctuation, and comments.

☆ **Compiled Languages** - You write the code, then a special program called a

"compiler" translates the code to "machine language" all at once.

☆ **Interpreted Language** - another program called an "interpreter" translates your code into machine language line by line as the program runs.

☆ **Assembly Language** - It is human readable, and it can interact directly with computer hardware. This is also known as a low-level language, as no compiler or interpreter is needed. It is hard to read, so it is not commonly used.

☆ **Structured Query Language (SQL)** - Query languages are used to interact with Databases. Queries can be used to search, add, or remove data from a database.

Planning a Coding Project

Defining the Goals

- ★ What will the software do?
- ★ How does the software use the hardware?
- ★ How will users interact with the software?
- ★ What will be displayed on the screen?

☆ **Flowchart** - Programmers use flowcharts to design the flow of their code.

☆ **Flowchart Shapes:**

- ★ Oval: The start or end of a program.
- ★ Parallelogram: Input or output.
- ★ Diamond: Decisions answered with a Yes or a No.
- ★ Rectangle: Processes.
- ★ Arrow: The logical flow of the program.

☆ **Pseudocode** - A way to describe how a computer program works using plain language instead of actual programming code.

☆ **Operators**

- ★ **==** means is *equal to*.
- ★ **!=** means is *not equal to*
- ★ **>** means *greater than*
- ★ **<** means *less than*
- ★ **>=** means *greater than or equal to*
- ★ **<=** means *less than or equal to*

☆ **Variables** - a container that can hold numbers, words, phrases, and other values. The value in a variable can vary or change.

Variable Types

☆ **Integers**: Positive or negative whole numbers.

☆ **Float**: Numbers with a decimal point.

☆ **Fixed-point**: Numbers with a fixed decimal point. (\$1.99, or \$456.78)

☆ **Double**: Similar to a float, but can store larger float numbers, or "double the precision."

☆ **Boolean**: Variable that stores only true or false.

☆ **String**: A sequence of characters such as "Hello World" and "555-1212."

☆ **Character**: A single character from any key on the keyboard: a letter, symbol, or digit.

☆ **Array**: A collection of variables of the same type, accessed with an index. An example would be like subjects in school = {Math, English, History, PE, Art, Science}.

☆ **Boolean and Logic** - Boolean operators compare if two values match.

☆ **Array** - A special type of variable that holds many values at once.

☆ **Vectors** - Groups of the same type of variable.

Booleans and Logic

☆ **AND** - If two conditions are true, the whole statement is true

☆ **OR** - If either condition is true, then the whole statement is true.

☆ **XOR** - If only one condition is true, the whole statement is true, but if both conditions are true, the statement is false

☆ **NOT** - Inverts a condition, true becomes false, and false becomes true.

Coding Principles

☆ **Program Flow** - the order a computer runs code, but that order can change based on various conditions.

☆ **Loops** - Programs can also have loops to repeat code while a condition exists.

☆ **Call Functions** - A Function Call flow happens when a program gets a 'call' to go to another part of the program, run the

code, and then return to where it was before.

☆ **Code Documentation** - Comments can be written with a special symbol in front of them such as #, ', or // to let the computer know to ignore them.

☆ **Branching** - Code can branch off and run different sets of code depending on conditions.

☆ **For Loops** - it is a control flow statement that repeats a block of code a specific number of times. It typically includes a starting point, an ending condition, and an increment/decrement step.

☆ **While Loops** - it is a control flow statement that repeats a block of code as long as a specified condition is true.

☆ **Functions** - A piece of code that does one specific job. It usually takes in data, processes it, and then gives a result.

Object-Orientated Programming

Object-Oriented Programming (OOP) -

A modern approach to programming. OOP Objects represent the specific model built from a Class (like a car or a bird). Objects can be programmed with variables, functions, or formulas.

Classes - It is like a blueprint or template for creating individual objects. Classes have properties, which are attributes that objects can have.

Objects - The specific models built from a class. The data inside objects can be variables of different types, including numbers, strings of text, and functions or formulas that can manipulate other data.

Troubleshooting and Debugging

- ★ Humans can make mistakes in coding, both with typing correctly and creating the proper code flow.
- ★ Troubleshooting code is running the program and fixing errors.
- ★ Troubleshooting and fixing code is known as debugging.
- ★ The first step in fixing errors is finding and repeating the problem.
- ★ Understanding the code is important

- ★ This is why writing comments in the code is important for good programming.
- ★ There are also debugging tools that help programmers interact with & analyze how the program behaves.
- ★ A divide and conquer approach can help focus on the problem and eventually find a solution.

